

AME 34351 - Check-in 1 - Group 5

1. Team Member Names and Planned Roles:

Michael Doyle - Visualization and Modeling Lead

John Carlson - Deployment/Communication

Peter Dacunto - Data Lead

Sophia Bracken - Problem/Domain Lead

2. Problem Statement

Our goal with this project is to predict the energy consumption of residential buildings using (approximately) eight architectural/geometric inputs: compactness, surface area, wall area, roof area, overall height, orientation, glazing area, and glazing area orientation. The goal is to minimize energy consumption for architects, engineers, and designers who want to create new and energy-efficient building designs.

3. Why This Problem Matters

In today's age, where non-renewable energy sources are becoming more and more scarce, it is important to minimize energy consumption in every possible way. While the easiest method would simply be to find better energy sources, in cases where non-renewable energy must be used, it's important to design and engineer buildings that will consume as little energy as possible.

4. Solution Benefits

There are many different interest groups who would benefit from this solution, including architects and engineers focused on designing energy-efficient buildings. In addition, private and public developers would benefit from more energy-efficient building designs, as this would drive overall costs down. Finally, with lower costs for developers and landowners, building residents also benefit.

5. Description of ML Task (Type)

This problem will be a Supervised Regression machine learning task; we know how different data points for building designs will correspond to energy consumption values, and we're aiming to determine the relationships between those data points using a regression model.

6. Data Sources

Commercial Data Energy Consumption Survey:

https://catalog.data.gov/dataset/commercial-buildings-energy-consumption-survey?from_hint=eyJxIjoiaZW5lcmd5IGNvbnN1bXB0aW9uIGluIGJ1aWxkaW5ncyIsInNvcnQiOiJwb3B1bGFyaXR5Ino%3D

Consumption and Efficiency Data and Statistics:

https://catalog.data.gov/dataset/consumption-efficiency-data-and-statistics?from_hint=eyJxIjoizW5lcmd5IGNvbN1bXBoaW9uIGluIGJ1aWxkaW5ncyIsInNvcnQiOiJwb3B1bGFyaXR5Ino%3D

7. Machine Learning Baseline for the Problem

A simple baseline would be to use the heat transfer calculation given by:

$$Q = U * A * \Delta T$$

Where:

Q = heat transfer (Watts)

U = heat transfer coefficient

A = surface area (units²)

ΔT = inside and outside temperature difference (F or C)

This equation is merely a baseline; this formula does not account for many architectural nuances in buildings, especially window surface area and various structural orientations.

Ultimately, the primary task of the model will be to determine how (and how much) the different design characteristics affect the energy consumption of each building.

Our research and calculations will help us to determine the weights ($w_1, w_2, \dots, w_l$, based on the calculations described above as well as any other formulas we discover) to assign to each data point ($x_1, x_2, \dots, x_l$, with each data point corresponding to a particular characteristic of each building) for the regression model.